Python Practice Questions with Answers

1. Find the Largest Number in a List

Question: Take a list of numbers and find the largest number.

Answer

```
numbers = [12, 45, 8, 67, 23]

largest = numbers[0]

for num in numbers:
    if num > largest:
        largest = num

print("Largest:", largest)
```

2. Count Even Numbers in a List

Question: Count how many even numbers are present in a list.

Answer

```
numbers = [1, 2, 4, 7, 8, 10, 13]

count = 0

for num in numbers:
    if num % 2 == 0:
        count += 1

print(count)
```

3. Reverse a List Using Slicing

Question: Reverse a list without using reverse().

Answer

```
numbers = [10, 20, 30, 40, 50]

reversed_list = numbers[::-1]

print(reversed_list)
```

4. Find Length of a Tuple

Question: Find the number of elements in a tuple.

Answer

```
data = (10, 20, 30, 40, 50)

print(len(data))
```

5. Create Squares Using List Comprehension

Question: Create a list containing squares of numbers from 1 to 10.

Answer

```
squares = [x**2 for x in range(1, 11)]

print(squares)
```

6. Use Ternary Operator

Question: Print "Pass" if marks are 40 or above, otherwise print "Fail".

Answer

```
marks = 55

result = "Pass" if marks >= 40 else "Fail"

print(result)
```

7. Find Unique Elements

Question: Remove duplicate elements from a list.

Answer

```
numbers = [1, 2, 2, 3, 4, 4, 5]

unique = list(set(numbers))

print(unique)
```

8. Count Frequency of Each Element

Question: Count how many times each element appears in a list.

Answer

```
numbers = [1, 2, 2, 3, 1, 4, 2]

freq = {}

for num in numbers:
  freq[num] = freq.get(num, 0) + 1

print(freq)
```

9. Find Common Elements Between Two Lists

Answer

```
list1 = [1, 2, 3, 4, 5]
list2 = [4, 5, 6, 7]

common = []

for item in list1:
  if item in list2:
    common.append(item)

print(common)
```

10. Separate Even and Odd Numbers

Answer

```
numbers = [1, 2, 3, 4, 5, 6, 7, 8]

even = []
odd = []

for num in numbers:
    if num % 2 == 0:
        even.append(num)
    else:
        odd.append(num)

print("Even:", even)
print("Odd:", odd)
```

11. Merge Two Dictionaries

Answer

```
d1 = {"a": 1, "b": 2}
d2 = {"c": 3, "d": 4}

d1.update(d2)

print(d1)
```

12. Find Students with Marks Above 75

Answer

```
marks = {
    "Aman": 82,
    "Riya": 65,
```

```
    "Raj": 91,
    "Priya": 74
}

for student, score in marks.items():
    if score > 75:
        print(student)
```

13. Create List of Even Numbers Using List Comprehension

Answer

```
evens = [x for x in range(1, 51) if x % 2 == 0]

print(evens)
```

14. Find Second Largest Number

Answer

```
numbers = [12, 45, 8, 67, 23, 55]

numbers.sort()

print(numbers[-2])
```

15. Find Missing Number

Question: A list contains numbers from 1 to 10 with one number missing. Find the missing number.

Answer

```
numbers = [1, 2, 3, 4, 5, 7, 8, 9, 10]

for i in range(1, 11):
    if i not in numbers:
        print("Missing:", i)
```

16. Word Frequency Counter

Answer

```
sentence = "python is easy and python is powerful"

words = sentence.split()

freq = {}

for word in words:
    freq[word] = freq.get(word, 0) + 1

print(freq)
```

17. Find Duplicate Elements

Answer

```
numbers = [1, 2, 3, 2, 4, 5, 1, 6]

duplicates = set()

for num in numbers:
    if numbers.count(num) > 1:
        duplicates.add(num)

print(duplicates)
```

18. Matrix Row Sum

Question: Find the sum of each row in a 2D list.

Answer

```
matrix = [
    [1, 2, 3],
    [4, 5, 6],
    [7, 8, 9]
]
```

```
for row in matrix:
    print(sum(row))
```

19. Find Dictionary Key with Highest Value

Answer

```
scores = {
    "Aman": 85,
    "Riya": 92,
    "Raj": 88
}

highest = max(scores, key=scores.get)

print(highest)
```

20. Set Operations

Question: Find union, intersection, and difference of two sets.

Answer

```
A = {1, 2, 3, 4, 5}
B = {4, 5, 6, 7}

print("Union:", A | B)
print("Intersection:", A & B)
print("Difference:", A - B)
```